

Supplier :		Keyence Singapore	Date:	01-Feb-17	Time: 09:30
Participant From Maruwa		Participant From Organizer	Report To		
1. Mr Siva	MM QA	1. Mr Takeshi	Regional Manger		
2.			1. Mr.Anthony,Mr.Ogawa		
3.			2. Mr.Seeni,Mr.Naga,Mr.Sinna		
4.			3. Mr.Theva,Mr.Shahrudin,Mr.Fauzi		
5.			4 Mr Sathia, Mr Fauzi		
			5.		

Agenda

Testing starts at 930am
 Equipment model : Keyence VR -3000 series
 Purpose :

DEPT	Measurement Item
CV	Flatness, Roughness & TA
Substrate	T.H. Shift, Edge Warp, Warp, Crack, Inner Dimension etc
TPH	Spot bumpy, scar

Below is the summary of the evaluation

	INSPECTION	JUDGEMENT	COMMENT	Propose
CV	Flatness	NG	Not able to calculate as a total but reference to point	-
	Roughness	OK	Measurable but need tuning on the setting	
SUBSTRATE DEPARTMENT	Thru Hole Shifted	NG	Not able to detect slitline for reference	Keyence IM model
	Edge Warp	OK	Can be measured. Tested on SA-10A NG product => Value 51um	
	Crack	NG	Can't measure thru profile	
	Inner Diamension	NG	Not able to detect slitline for reference	Keyence IM model
	Shedder break	NG	Can measure	
	Slitline Depth	NG	Can't measure (Wrong value)	
	Warp	OK	Can measure=> judgement value >100um	4min/pc
	Roughness	OK	OK	
TPH	Bumpy	NG	Compare with surface roughness machine, the measurement value smaller. Result Surface roughness: 60um Keyence 20um. The machine calculating base from surface area average	
	Dent	NG	Didn't test	

Meeting adjourned at 12:30pm

Conclusion & Opinion

This equipment capability is high and application towards MARUWA product is very minimal. The ROI vs manpower reduction is low but highly efficient especially on the data collection. Will further evaluate other machine in order to cater MARUWA needs.

KEYENCE EQUIPMENT EVALUATION RESULT

01st Feb 2017

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